

Abdullah Mohammad Aref

Berlin, Germany . abdullahmohammadaref@gmail.com

linkedin.com/in/abdullahmohammadaref | github.com/Abdullahmohammadaref

Professional Summary

Versatile data scientist, backend developer, and 5th-semester Computer Science student at Gisma University of Applied Sciences specializing in predictive modelling, machine learning algorithms, database management, and RESTful APIs. Experienced in utilizing Python, R, NumPy, and Scikit-Learn to leverage datasets and engineer insightful machine learning models and, Django, Node.js, Express.js, TypeScript, MongoDB, and SQLite3 to create secure and stable backend applications. A data-driven problem solver striving to build scalable software solutions that address real-world problems and provide actionable data insights.

Education

Gisma University of Applied Sciences - BSc Computer Science Berlin, Germany 2023 – Current

- Relevant Modules: Artificial Intelligence & Machine Learning, Applied Statistical Modeling, Software Engineering, Data Structures & Algorithms, Cyber Security, App & Web Development, Databases & Big Data.

Experience

Freelance Software Engineer - Berlin, Germany 09/2025 – Current

- Rebuilt legacy Microsoft Access vehicle management system into a modern Django and React application. Built an advanced database to store data and automate administrative tasks.
- Implemented secure authorization and 2-factor authentication using one-time tokens. Enhanced user experience with pagination, internationalization, and autosaving forms.
- Leveraged cutting-edge AI tools for rapid development, and deployed the application using Docker, then hosted it on an Oracle Cloud instance with GitHub CI/CD pipelines.

Projects

Flow (Codebase Mapping Tool) | Python, React, Flask, Node.js, Typescript 04/2026 – Current (Till 06/2026)

- Built an end-to-end codebase visualizer, leveraging full-stack development with machine learning to facilitate developer onboarding and quickly map unfamiliar projects.
- Engineering a supervised machine learning pipeline that currently trains on 240 files and classifies them based on their role with a 76% macro F1-score.

House Pricing Estimator | Python, Pandas, Scikit-Learn: 07/2025 – 09/2025

- Built an end-to-end machine learning regression analysis pipeline in Python to estimate housing prices with high accuracy, utilizing the Pandas library for transforming and cleaning data, and NumPy, Seaborn, and Matplotlib for data exploration and plotting.
- Engineered features using Scikit-Learn's ColumnTransformer with OneHotEncoder and StandardScaler to optimize numerical and categorical data efficiently
- Optimized models with hyperparameter tuning using RandomizedSearchCV and K-fold cross-validation to ensure reliable results.
- Achieved a high R2 score using a Random Forest Regressor model, explaining 83.1% of factors that affect housing prices.

Telecom Customer Churn Prediction | R, Logistic Regression: 07/2025 – 09/2025

- Conducted a fully-fledged statistical analysis using R to predict customer churn, utilizing ggplot2 and ggcorrplot to explore and visualize data.
- Applied data preprocessing, sampling, and cleaning with methods like scaling numerical data to reduce the effect of outliers.
- Engineered hypotheses and tested them using Chi-squared tests for categorical variables and Binomial Logistic Regression for continuous variables.
- Optimized the model results by calculating the Variance Inflation Factor (VIF) to detect and remove multicollinearity, utilized the ROC curve and AUC metrics to evaluate and enhance the model's predictive power by maximizing recall to 94.7% in order to effectively capture potential churning customers.

E-commerce Web Application | Vue.js, Express.js, Node.js, MongoDB, JavaScript 01/2026 – 03/2026

- Developed a full-stack e-commerce store with the MEVN stack featuring a responsive user interface with CSS Flexbox, Grid, and media queries.
- Built RESTful APIs to support CRUD operations, implementing Secure authentication, authorization, and access control using Bcrypt to hash and salt passwords.
- Integrated PayPal Sandbox API to support payments, managed user and cart state with Pinia, and optimized image processing with Multer and Sharp

Password Manager | Python, Django, SQLite3, Cryptography

04/2025 – 06/2025

- Programmed a password manager web application using the Django framework, allowing users to securely store, generate, edit, and view their account credentials.
- Implemented data protection by hashing and salting master passwords and encrypting stored credentials using the Cryptography library.
- Developed a secure authentication system using email Two-Factor Authentication (2FA), and temporary tokens.
- Protected from web attacks by utilizing Django's Object-Relational Mapping (ORM) to prevent SQL injections, token validation to block CSRF attacks, and auto-escaping to stop XSS threats.

Technical & Soft Skills

- Programming Languages: Python, Java, JavaScript/Typescript, R, HTML, CSS, SCSS, LaTeX, Markdown.
- Frameworks & Libraries: Scikit-Learn, React, Vue.js, Next.js, Node.js, Express.js, Nest.js, Django, Bootstrap, Tailwind CSS, NumPy, Matplotlib, Seaborn, Pandas.
- Databases: SQLite3, MySQL, MongoDB.
- Tools & Cloud Environments: Git, Jupyter Notebook, Oracle & AWS Cloud, Docker, AI development tools
- Core Concepts: Machine Learning, Predictive modeling, Data structures & Algorithms, Concurrency (Multithreading), Object-Oriented Programming (OOP), Cybersecurity (Firewalls, Encryption, Hashing, Salting, 2FA, Authentication, Authorization), Unit testing.
- Soft Skills: Adaptability to changes, Agile learning (Growth Mindset), Target-Oriented Problem Solving, and Teamwork.

Languages

English - C1 | German – A2 | Arabic – Native